



Using R on Oscar

Joselynn Wallace, PhD
Lead Genomics Data Scientist
COBRE Computational Biology Core
Center for Computation and Visualization



Goals

- Why use R?
- Options for using R on Oscar
- How to find and install R packages
- Debugging R package installations
- Reproducibility and containerization of R analyses



Why use R?

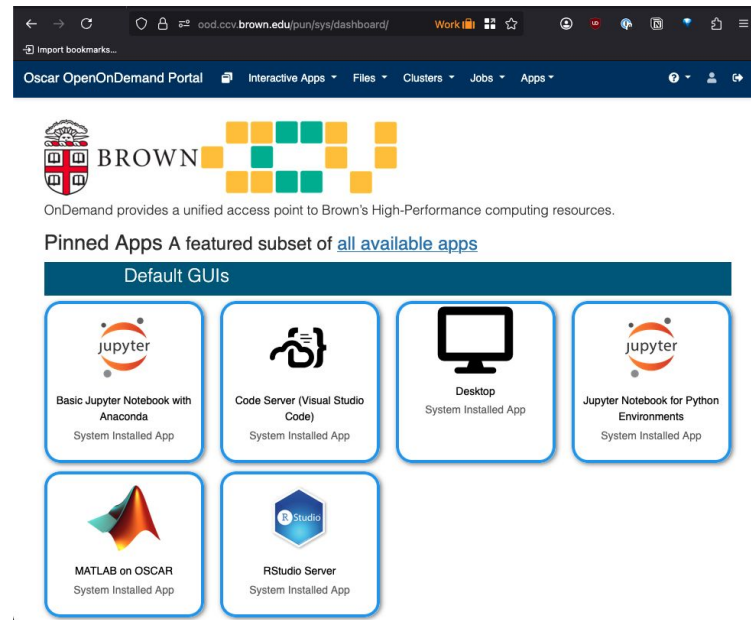
- R is a language and environment for statistical computing and graphics (<https://www.r-project.org/about.html>).
- You can use R interactively (e.g., enter commands into the R console or R Studio <https://rstudio.com/>).
- You can write R scripts and run them from the terminal or generate reports and documents using R notebooks (<https://blog.rstudio.com/2016/10/05/r-notebooks/>), markdown files (<https://rmarkdown.rstudio.com/>), or Quarto (<https://quarto.org/>).
- It is free and can be tailored to your needs by installing and using specific packages from places like CRAN (<https://cran.r-project.org/>), BioConductor (<https://www.bioconductor.org/>), or GitHub.



Options for using R on Oscar

Using R with Open On Demand

- Navigate to ood.ccv.brown.edu
- Authenticate using Shibboleth/your Brown ID
- **RStudio**
 - Default GUIs > RStudio Server
 - Expert GUIs > RStudio Singularity
- **R console**
 - Expert GUIs > OSCAR Shell Access





Using R with Terminal

- After connecting to Oscar (SSH on terminal or OOD OSCAR Shell Access), use `module spider r` to see all R modules available.
- Load the specific version you want (`module load r/4.5.1-iikl`), then type `R` to start console.

```
[jwalla12@login010 ~]$ module load r/4.5.1-iikl
[jwalla12@login010 ~]$ R
```

```
R version 4.5.1 (2025-06-13) -- "Great Square Root"
Copyright (C) 2025 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu
```

```
R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.
```

```
  Natural language support but running in an English locale
```

```
R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.
```

```
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.
```

```
> █
```



How to find R packages



Finding R packages on CRAN

- CRAN has a list of all available packages (https://cran.r-project.org/web/packages/available_packages_by_name.html)
- Also has Task Views page with packages organized by topic: <https://cran.r-project.org/web/views/>.
- Install new packages using `install.packages()` function.
- **Note some packages will require code to be compiled so it is best to do R package installs on the login node.**

```
[jwalla12@login010 ~]$ module load r/4.5.1-iiikl
[jwalla12@login010 ~]$ R

R version 4.5.1 (2025-06-13) -- "Great Square Root"
Copyright (C) 2025 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

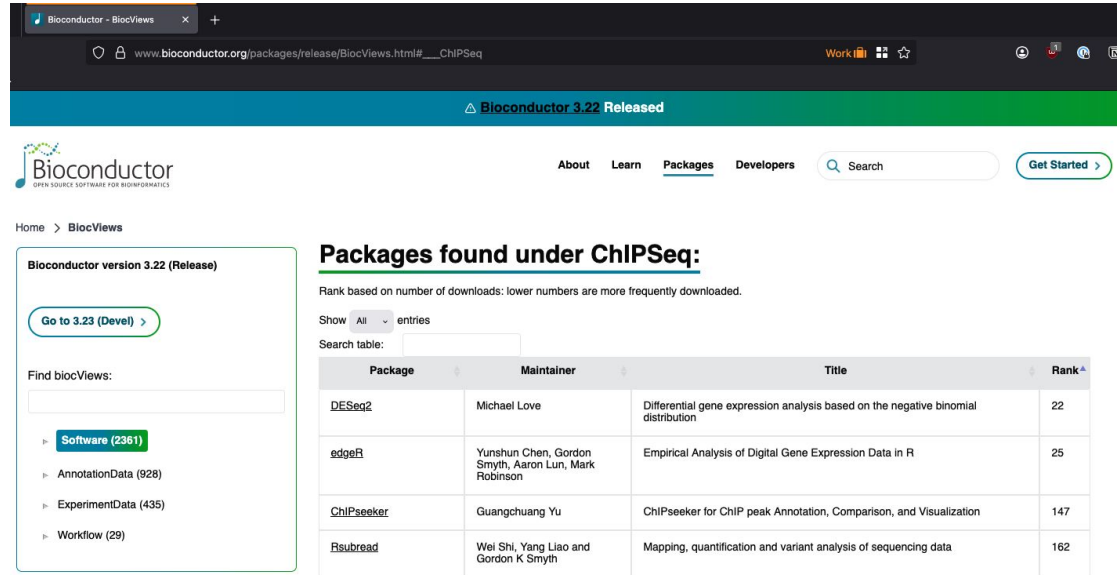
R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> install.packages('sf')
```


Finding R packages on Bioconductor

- Bioconductor is another repository of R packages, similar to CRAN, but focuses on biological data.
- You can see a list of all packages here: https://www.bioconductor.org/packages/release/BiocViews.html#___Software, which you can also filter to view packages organized by specific task or topic.
- To install bioconductor packages, install BiocManager from CRAN
`install.packages("BiocManager")`.
- Then use `BiocManager::install()` to install specific packages.



The screenshot shows the Bioconductor website interface. At the top, there's a navigation bar with links for 'About', 'Learn', 'Packages', and 'Developers'. A search bar is also present. Below the navigation bar, the 'Bioconductor version 3.22 (Release)' is indicated. The main content area is titled 'Packages found under ChIPSeq:' and displays a table of packages. The table has columns for 'Package', 'Maintainer', 'Title', and 'Rank'. The packages listed are DESeq2, edgeR, ChIPseeker, and Rsubread. The rank is based on the number of downloads, with lower numbers indicating more frequent downloads.

Bioconductor version 3.22 (Release)

[Go to 3.23 \(Devel\)](#)

Find bioViews:

- [Software \(2361\)](#)
- [AnnotationData \(928\)](#)
- [ExperimentData \(435\)](#)
- [Workflow \(29\)](#)

Packages found under ChIPSeq:

Rank based on number of downloads: lower numbers are more frequently downloaded.

Show [All](#) entries

Search table:

Package	Maintainer	Title	Rank
DESeq2	Michael Love	Differential gene expression analysis based on the negative binomial distribution	22
edgeR	Yunshun Chen, Gordon Smyth, Aaron Lun, Mark Robinson	Empirical Analysis of Digital Gene Expression Data in R	25
ChIPseeker	Guangchuang Yu	ChIPseeker for ChIP peak Annotation, Comparison, and Visualization	147
Rsubread	Wei Shi, Yang Liao and Gordon K Smyth	Mapping, quantification and variant analysis of sequencing data	162



How to install R packages



Installing R packages

- Install packages from CRAN
(https://cran.r-project.org/web/packages/available_packages_by_name.html)
using `install.packages()`
- Install packages from Bioconductor
(https://www.bioconductor.org/packages/release/BiocViews.html#___Software) using `BiocManager::install()`
- If the packages you want aren't in either of these places, you can install from GitHub using `install_github()` function from either remotes <https://remotes.r-lib.org/> or devtools <https://devtools.r-lib.org/> packages.
- Again, some packages will require code to be compiled so it is best to do R package installs on the login node.

Debugging R packages installs

- I tried running
`install.packages('sf')`
and got an error message
stating that the installation
had non-zero exit status.

```
jwalla12@login010:~  
** help  
*** installing help indices  
** building package indices  
** installing vignettes  
** testing if installed package can be loaded from temporary loc  
ation  
** checking absolute paths in shared objects and dynamic librari  
es  
** testing if installed package can be loaded from final locatio  
n  
** testing if installed package keeps a record of temporary inst  
allation path  
* DONE (classInt)  
ERROR: dependency 'units' is not available for package 'sf'  
Perhaps try a variation of:  
install.packages('units')  
* removing '/oscar/home/jwalla12/R/x86_64-pc-linux-gnu-library/4  
.5/sf'  
  
The downloaded source packages are in  
  '/tmp/RtmpVlxu4k/downloaded_packages'  
Warning messages:  
1: In install.packages("sf") :  
  installation of package 'units' had non-zero exit status  
2: In install.packages("sf") :  
  installation of package 'sf' had non-zero exit status  
> 
```



Debugging R packages installs

- To debug, I scroll back up in the terminal history to find this error message about libudunits2.so not being found and another message about gdal-config not being found.

```
-----  
Configuration failed because libudunits2.so was not found. Try installing:  
* deb: libudunits2-dev (Debian, Ubuntu, ...)  
* rpm: udunits2-devel (Fedora, EPEL, ...)  
* brew: udunits (OSX)  
If udunits2 is already installed in a non-standard location, use:  
--configure-args='--with-udunits2-lib=/usr/local/lib'  
if the library was not found, and/or:  
--configure-args='--with-udunits2-include=/usr/include/udunits2'  
if the header was not found, replacing paths with appropriate values.  
You can alternatively set UDUNITS2_INCLUDE and UDUNITS2_LIBS manually.  
-----
```

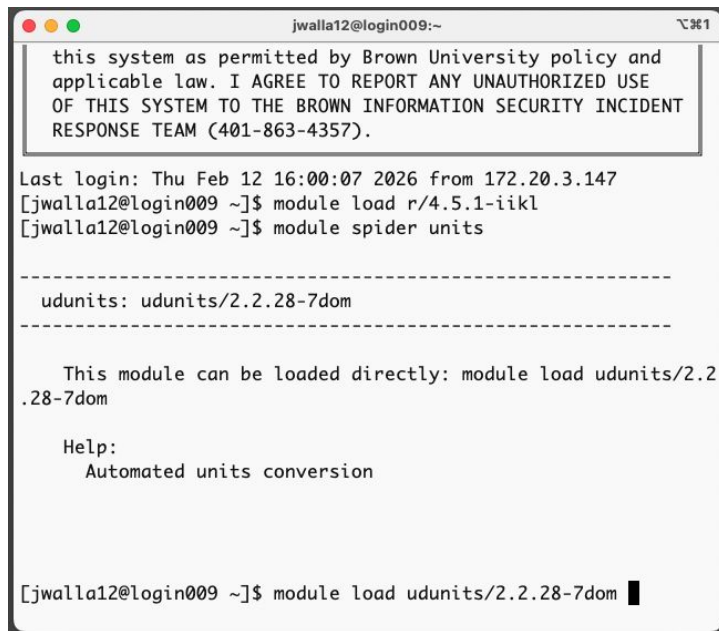
```
See 'config.log' for more details  
ERROR: configuration failed for package 'units'  
* removing '/oscar/home/jwalla12/R/x86_64-pc-linux-gnu-library/4.5/units'
```

```
...  
configure: error: gdal-config not found or not executable.  
*** Installing this package from source requires the prior  
*** installation of external software, see for details  
*** https://r-spatial.github.io/sf/#installing  
ERROR: configuration failed for package 'sf'  
* removing '/oscar/home/jwalla12/R/x86_64-pc-linux-gnu-library/4.5/sf'
```

```
The downloaded source packages are in  
  '/tmp/RtmpWA73cS/downloaded_packages'  
Warning message:  
In install.packages("sf") :  
  installation of package 'sf' had non-zero exit status  
> █
```

Debugging R packages installs

- I use `module spider` `units` and `module spider gdal` to see which modules are available and load them before trying to reinstall the R package.

A terminal window titled 'jwalla12@login009:~' with standard macOS window controls (red, yellow, green buttons) in the top-left corner. The terminal displays a system policy notice in a box, followed by login information and two 'module load' commands. It then shows the output of 'module spider units', which lists 'udunits: udunits/2.2.28-7dom'. Below this, it provides instructions on how to load the module directly and offers help for 'Automated units conversion'. The final line shows the command 'module load udunits/2.2.28-7dom' being entered.

```
jwalla12@login009:~  
this system as permitted by Brown University policy and  
applicable law. I AGREE TO REPORT ANY UNAUTHORIZED USE  
OF THIS SYSTEM TO THE BROWN INFORMATION SECURITY INCIDENT  
RESPONSE TEAM (401-863-4357).  
  
Last login: Thu Feb 12 16:00:07 2026 from 172.20.3.147  
[jwalla12@login009 ~]$ module load r/4.5.1-iiikl  
[jwalla12@login009 ~]$ module spider units  
  
-----  
udunits: udunits/2.2.28-7dom  
-----  
  
This module can be loaded directly: module load udunits/2.2.  
.28-7dom  
  
Help:  
Automated units conversion  
  
[jwalla12@login009 ~]$ module load udunits/2.2.28-7dom
```

Debugging R packages installs

- However, still seeing error messages about libproj and sqlite3 not found in the standard or given locations.

```
jwalla12@login009:~  
checking for gdal.h... yes  
checking GDAL: linking with --libs only... yes  
checking GDAL: /oscar/rt/9.6/25/spack/x86_64_v3/gdal-3.11.0-4d3c37lafseqzfi77nibzgcwasycjs/share/gdal/pcs.csv readable... no  
checking GDAL: checking whether PROJ is available for linking:... yes  
checking GDAL: checking whether PROJ is available for running:... yes  
configure: GDAL: 3.11.0  
configure: pkg-config proj exists, will use it  
configure: using proj.h.  
configure: PROJ: 9.4.1  
checking PROJ: checking whether PROJ and sqlite3 are available for linking:... no  
configure: error: libproj or sqlite3 not found in standard or given locations.  
*** Installing this package from source requires the prior  
*** installation of external software, see for details  
*** https://r-spatial.github.io/sf/#installing  
ERROR: configuration failed for package 'sf'  
* removing '/oscar/home/jwalla12/R/x86_64-pc-linux-gnu-library/4.5/sf'  
  
The downloaded source packages are in  
  '/tmp/RtmprqEPdo/downloaded_packages'  
Warning message:  
In install.packages("sf") :  
  installation of package 'sf' had non-zero exit status  
>
```



Debugging R packages installs

- Load the necessary modules (gdal, udunits, proj, and sqlite).
- When running `install.packages()`, specify paths to proj and sqlite3.
- See the Oscar docs here for more information: <https://docs.ccv.brown.edu/oscar/software/r-installs>

```
Last login: Thu Feb 26 10:12:27 2026 from 172.20.3.203
[jwalla12@login010 ~]$ module load gdal/3.11.0-4d3c
[jwalla12@login010 ~]$ module load udunits/2.2.28-7dom
[jwalla12@login010 ~]$ module load proj/9.4.1-fhhv
[jwalla12@login010 ~]$ module load sqlite/3.46.0-pwg7
[jwalla12@login010 ~]$ module load r/4.5.1-iiikl
[jwalla12@login010 ~]$ R
```

```
R version 4.5.1 (2025-06-13) -- "Great Square Root"
Copyright (C) 2025 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu
```

```
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Type 'licence()' or 'licence()' for distribution details.
```

```
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```

```
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.
```

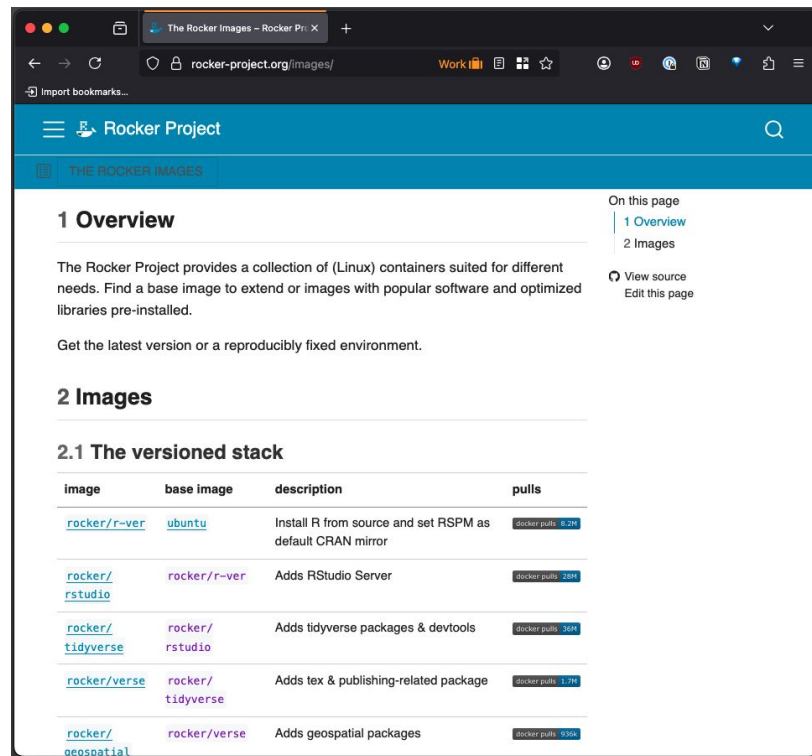
```
> install.packages("sf", configure.args = c("--with-proj-include=
/oscar/rt/9.6/25/spack/x86_64_v3/proj-9.4.1-fhhvfr6iwri5cw42roir5
meojvazzfoo/include", "--with-proj-lib=/oscar/rt/9.6/25/spack/x86
_64_v3/proj-9.4.1-fhhvfr6iwri5cw42roir5meojvazzfoo/lib64", "--wit
h-sqlite3-lib=/oscar/rt/9.6/25/spack/x86_64_v3/sqlite-3.46.0-pwg7
tka36sc5abuw7l637zgwrvk63tu3/lib"))
```




Reproducibility and containerization of R analyses

Rocker/Singularity RStudio

- Containers allow you to restrict your computing environment to a specific version of R and R packages.
- You can share your computing environment with others by giving them access to the Dockerfile or Docker image, which makes your analyses more reproducible.
- Rocker project (<https://rocker-project.org/>) organized a collection of Docker images to run containerized instances of RStudio that can run as Singularity images on Oscar.
- You can find the base images on Dockerhub <https://hub.docker.com/u/rocker>.
- The relevant Docker files are on their github repository <https://github.com/rocker-org/rocker-versioned2>.
- Can use pre-built Rocker containers as-is or build on top of them using custom Docker files.



The screenshot shows the Rocker Project website. The page has a blue header with the "Rocker Project" logo and a search icon. Below the header, there's a section titled "THE ROCKER IMAGES". The main content area is divided into two columns. The left column contains the "Overview" and "Images" sections. The "Overview" section states that the Rocker Project provides a collection of (Linux) containers suited for different needs, and it provides a link to get the latest version or a reproducibly fixed environment. The "Images" section is titled "2 Images" and contains a sub-section "2.1 The versioned stack". This sub-section contains a table with four columns: "image", "base image", "description", and "pulls". The table lists several Docker images, including "rocker/r-ver", "rocker/rstudio", "rocker/tidyverse", "rocker/verse", and "rocker/geospatial". The right column of the page contains a "On this page" section with links to "1 Overview" and "2 Images", and a "View source" link.

image	base image	description	pulls
rocker/r-ver	ubuntu	Install R from source and set RSPM as default CRAN mirror	docker pulls: 8.4M
rocker/rstudio	rocker/r-ver	Adds RStudio Server	docker pulls: 28M
rocker/tidyverse	rocker/rstudio	Adds tidyverse packages & devtools	docker pulls: 36M
rocker/verse	rocker/tidyverse	Adds tex & publishing-related package	docker pulls: 1.1M
rocker/geospatial	rocker/verse	Adds geospatial packages	docker pulls: 930k



Using pre-built Rocker images

- Pull to oscar:

singularity pull <image_name>.sif docker://<dockerhub_user>/<name>:<tag>

```
[jwalla12@login010 r_workshop]$ singularity pull tidyverse:4.5.2.sif docker://rocker/tidyverse:4.5.2
INFO:   Environment variable SINGULARITY_CACHEDIR is set, but APPTAINER_CACHEDIR is preferred
INFO:   Converting OCI blobs to SIF format
INFO:   Starting build...
INFO:   Fetching OCI image...
294.4MiB / 294.4MiB [=====]
33.6MiB / 33.6MiB [=====]
28.3MiB / 28.3MiB [=====]
27.2MiB / 27.2MiB [=====]
717.2KiB / 717.2KiB [=====]
194.6MiB / 194.6MiB [=====]
344.2MiB / 344.2MiB [=====]
INFO:   Extracting OCI image...
INFO:   Inserting Apptainer configuration...
INFO:   Creating SIF file...
[=====]
[jwalla12@login010 r_workshop]$ ls
tidyverse:4.5.2.sif
[jwalla12@login010 r_workshop]$
```



Using pre-built Rocker images

- On OOD, navigate to Expert GUIs > RStudio Singularity
- Fill in the fields as in the image on the right, leaving the other fields empty, then scroll down and click the blue “Launch” button.

Number of hours

Num Cores

Indicate the number of cores you need. The default is 1

Memory (GB) (optional)

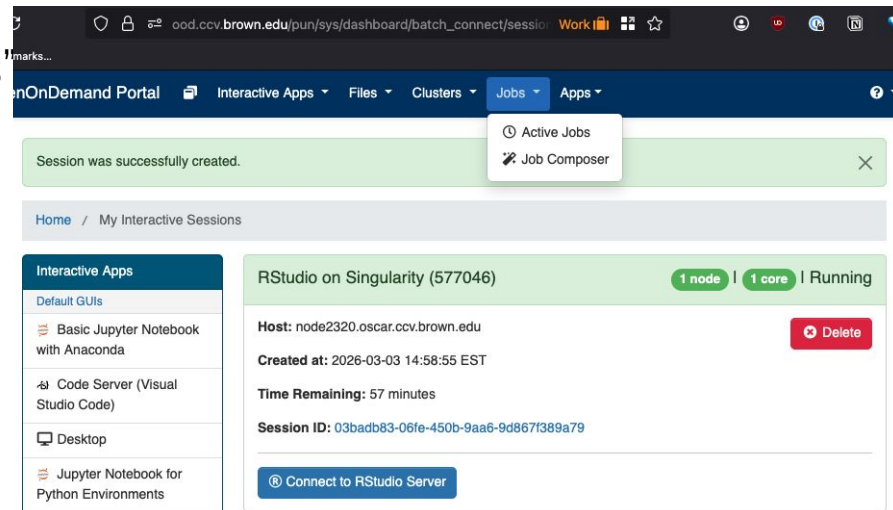
Amount of memory to allocate

Singularity Container Path (required)

The full path to the directory of the container that you wish you run. This will be the sif file generated when you use
`singularity pull docker://xyz`

Using pre-built Rocker images

- Click on “My Interactive Sessions”
- Click the blue button that says “Connect to RStudio Server”





Modifying Rocker Images

One potential workflow is:

1. Install Docker desktop (<https://www.docker.com/products/docker-desktop/>) and make a Dockerhub account
2. Write a dockerfile using Rocker images as the base image (<https://hub.docker.com/u/rocker>) and add install steps for your desired R packages.
3. Use buildx to build multi-platform images that will run on linux (Oscar) (<https://docs.docker.com/build/building/multi-platform/#build-multi-platform-images>), tagging and pushing to github or dockerhub container registries on build (probably need to login to Dockerhub <https://docs.docker.com/reference/cli/docker/login/>).
4. Pull images to Oscar using Singularity pull, proceed as you would for a pre-built Rocker image.



Example Dockerfile using a rocker tidyverse as base image

```
jwalla12@0IT2C4DQY3:~/chordoma_rna_example $ cat Dockerfile
FROM rocker/tidyverse:4.5.2

RUN apt-get update && apt-get install -y --no-install-recommends \
    libglpk-dev \
    && rm -rf /var/lib/apt/lists/*

RUN Rscript -e "\
install.packages(c('ashr', 'circlize', 'matrixStats', 'msigdb')); \
if (!requireNamespace('BiocManager', quietly = TRUE)) install.packages('BiocManager'); \
BiocManager::install(c( \
  'airway', \
  'DESeq2', \
  'ComplexHeatmap', \
  'EnhancedVolcano', \
  'clusterProfiler', \
  'org.Hs.eg.db', \
  'AnnotationDbi' \
), ask = FALSE) \
"
```



Build multi-platform image using buildx build

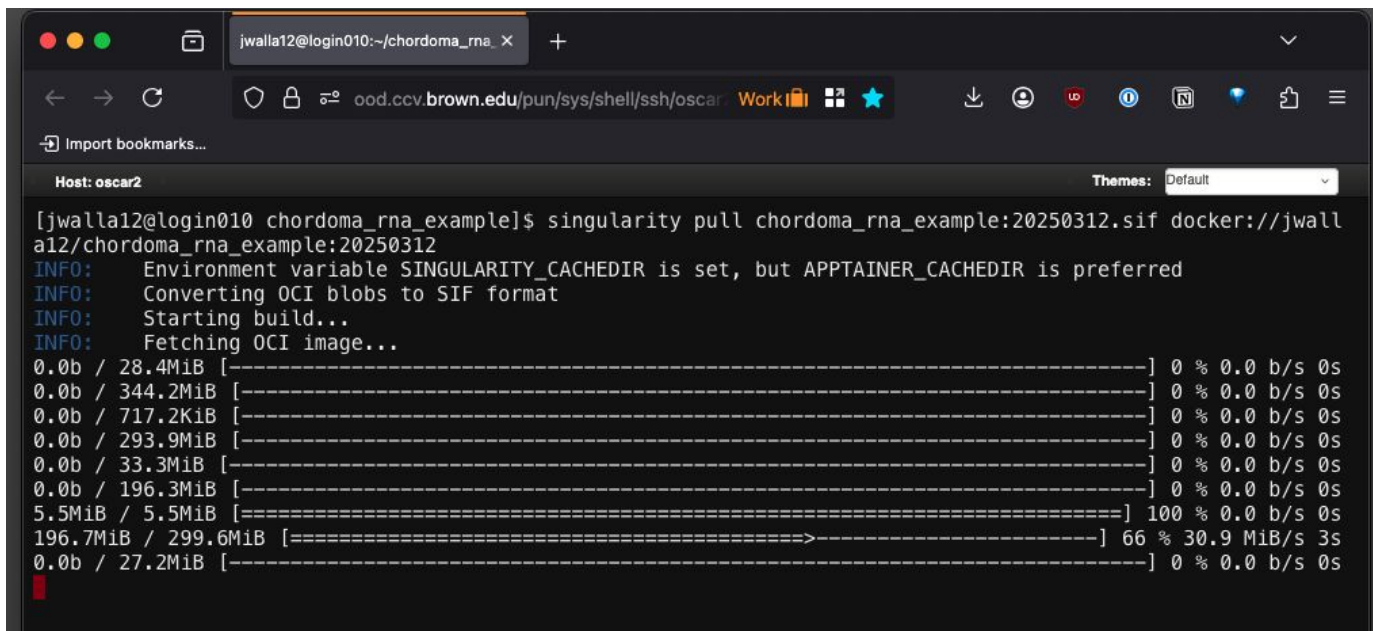
```
jwalla12@0IT2C4DQY3:~/chordoma_rna_example $ cat Dockerfile
FROM rocker/tidyverse:4.5.2

RUN apt-get update && apt-get install -y --no-install-recommends \
    libglpk-dev \
    && rm -rf /var/lib/apt/lists/*

RUN Rscript -e "\
install.packages(c('ashr', 'circlize', 'matrixStats', 'msigdb')); \
if (!requireNamespace('BiocManager', quietly = TRUE)) install.packages('BiocManager'); \
BiocManager::install(c( \
  'airway', \
  'DESeq2', \
  'ComplexHeatmap', \
  'EnhancedVolcano', \
  'clusterProfiler', \
  'org.Hs.eg.db', \
  'AnnotationDbi' \
), ask = FALSE) \
"

jwalla12@0IT2C4DQY3:~/chordoma_rna_example $ docker buildx build --platform linux/amd64,linux/arm64 --push -t jwalla12/chordoma_rna_example:20250312 .
```


Pull to Oscar using singularity pull



```
jwalla12@login010:~/chordoma_rna_ X
oood.ccv.brown.edu/pun/sys/shell/ssh/oscar: Work
Host: oscar2
[jwalla12@login010 chordoma_rna_example]$ singularity pull chordoma_rna_example:20250312.sif docker://jwalla12/chordoma_rna_example:20250312
INFO: Environment variable SINGULARITY_CACHEDIR is set, but APPTAINER_CACHEDIR is preferred
INFO: Converting OCI blobs to SIF format
INFO: Starting build...
INFO: Fetching OCI image...
0.0b / 28.4MiB [-----] 0 % 0.0 b/s 0s
0.0b / 344.2MiB [-----] 0 % 0.0 b/s 0s
0.0b / 717.2KiB [-----] 0 % 0.0 b/s 0s
0.0b / 293.9MiB [-----] 0 % 0.0 b/s 0s
0.0b / 33.3MiB [-----] 0 % 0.0 b/s 0s
0.0b / 196.3MiB [-----] 0 % 0.0 b/s 0s
5.5MiB / 5.5MiB [=====] 100 % 0.0 b/s 0s
196.7MiB / 299.6MiB [=====] 66 % 30.9 MiB/s 3s
0.0b / 27.2MiB [-----] 0 % 0.0 b/s 0s
```

Run on OOD using RStudio on Singularity

 RStudio Server

 Expert GUIs

 Desktop (Advanced)

 Jupyter Notebook for Apptainer Images

 RStudio on Singularity

Account

Partition

Number of hours

Num Cores

Indicate the number of cores you need. The default is 1

Memory (GB) (optional)

Amount of memory to allocate

Singularity Container Path (required)

The full path to the directory of the container that you wish you



Have Questions?

- Documentation : <https://docs.ccv.brown.edu/oscar/>
- General Oscar Support : support@ccv.brown.edu
- Office Hours : Fri 10 AM - 12PM (Providence time) via zoom
- Programming Help/Consultation : hpc@brown.edu

[CCV Events Calendar](#)